VLSI Devices
Lecture 3

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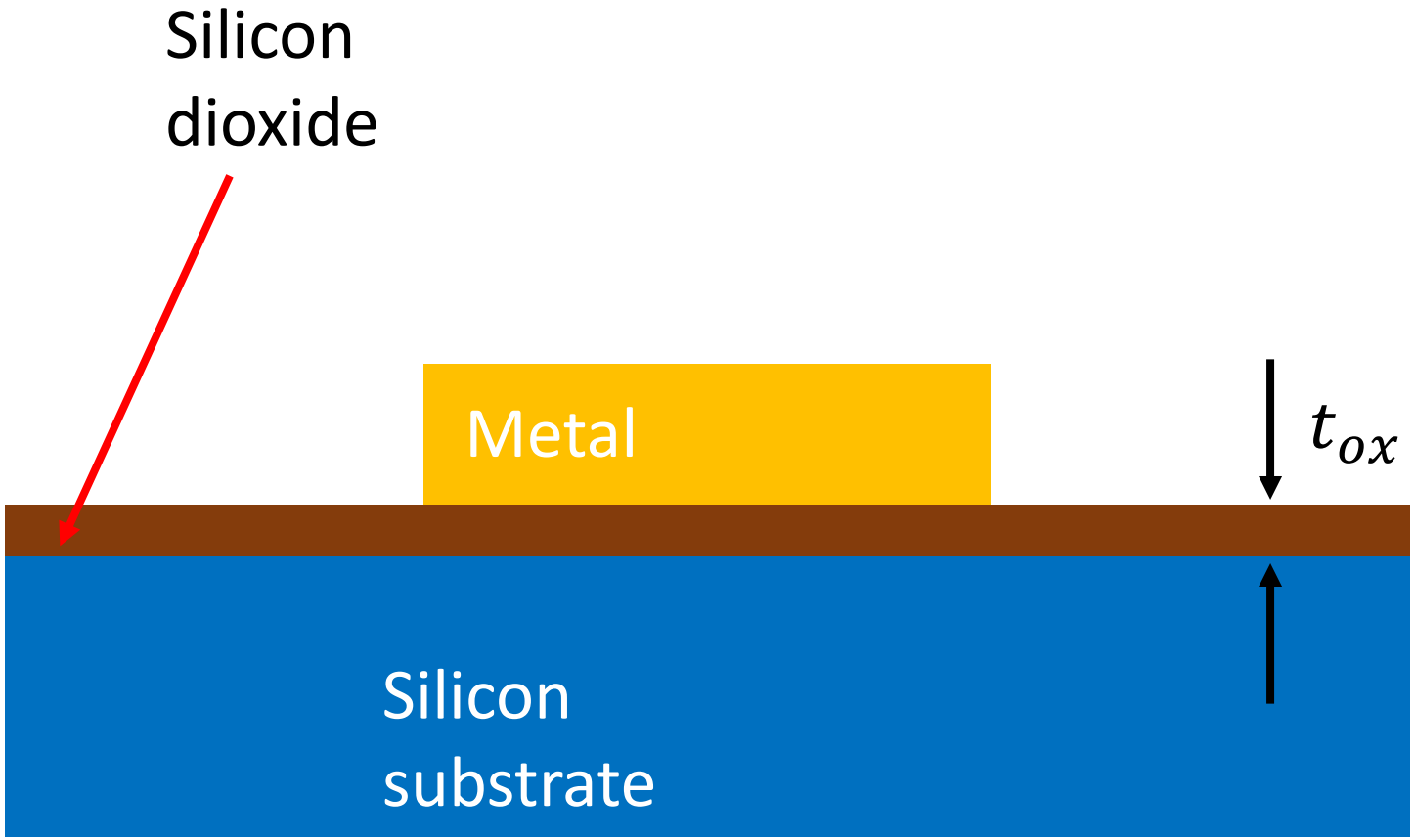
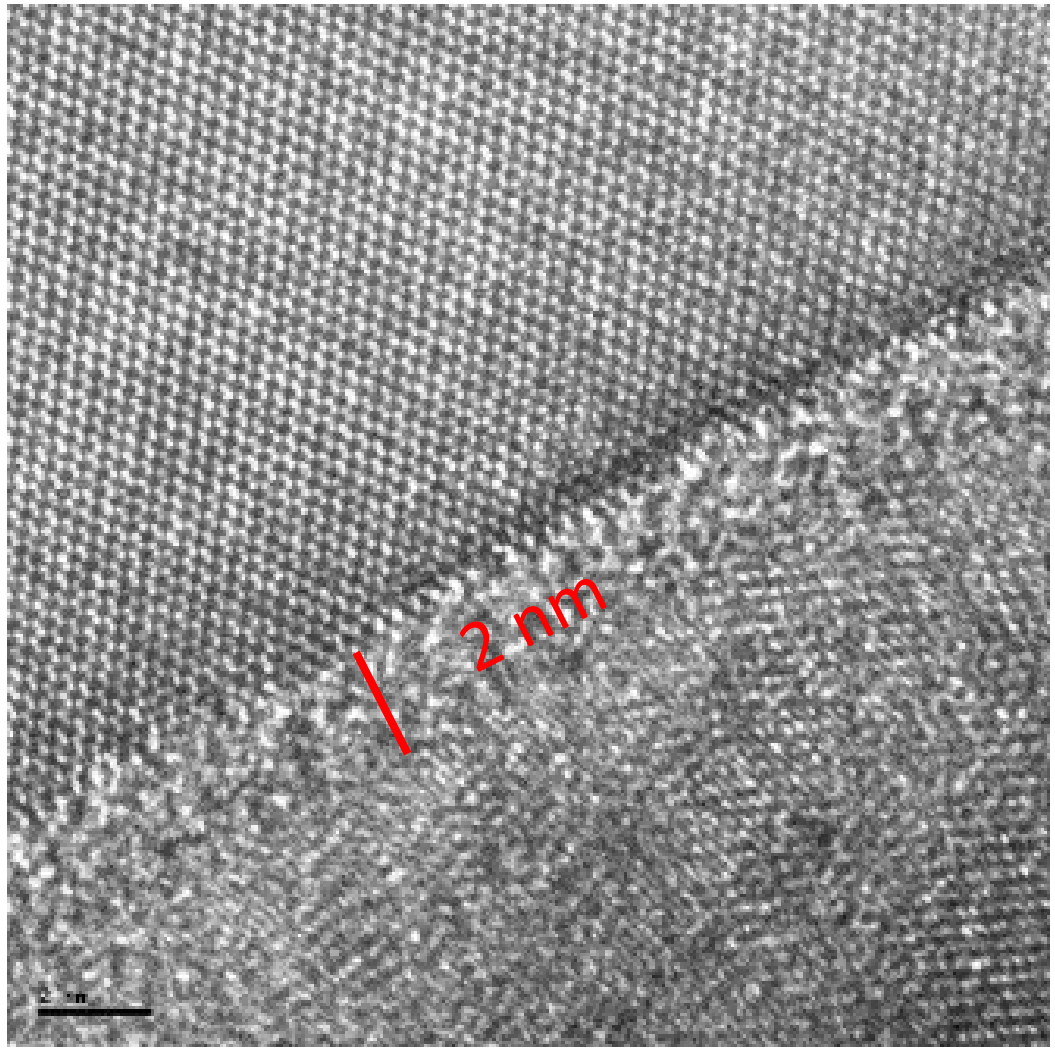
L3

MOS capacitors

- Basis of CMOS technology

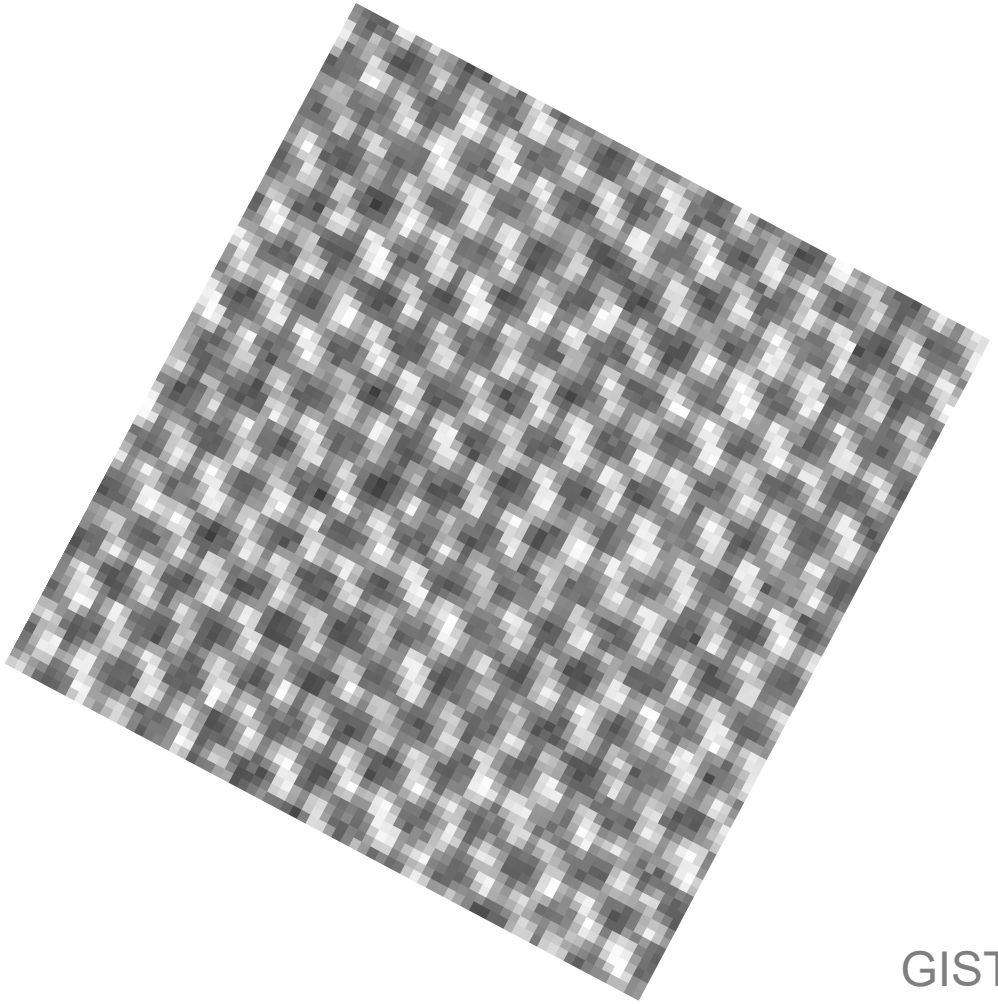


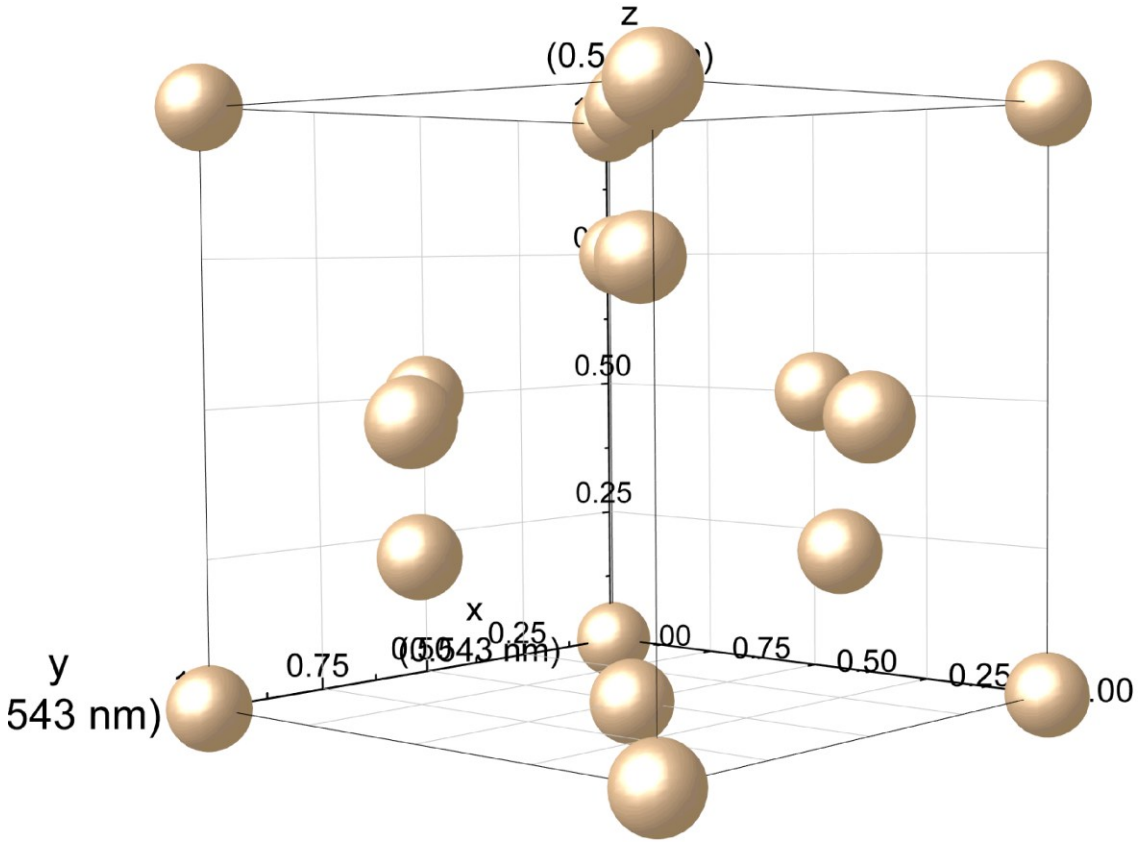


TEM image of a cross section of a MOSFET transistor (B. Freitag et al, Ultramicroscopy, 2005)

Crystalline silicon

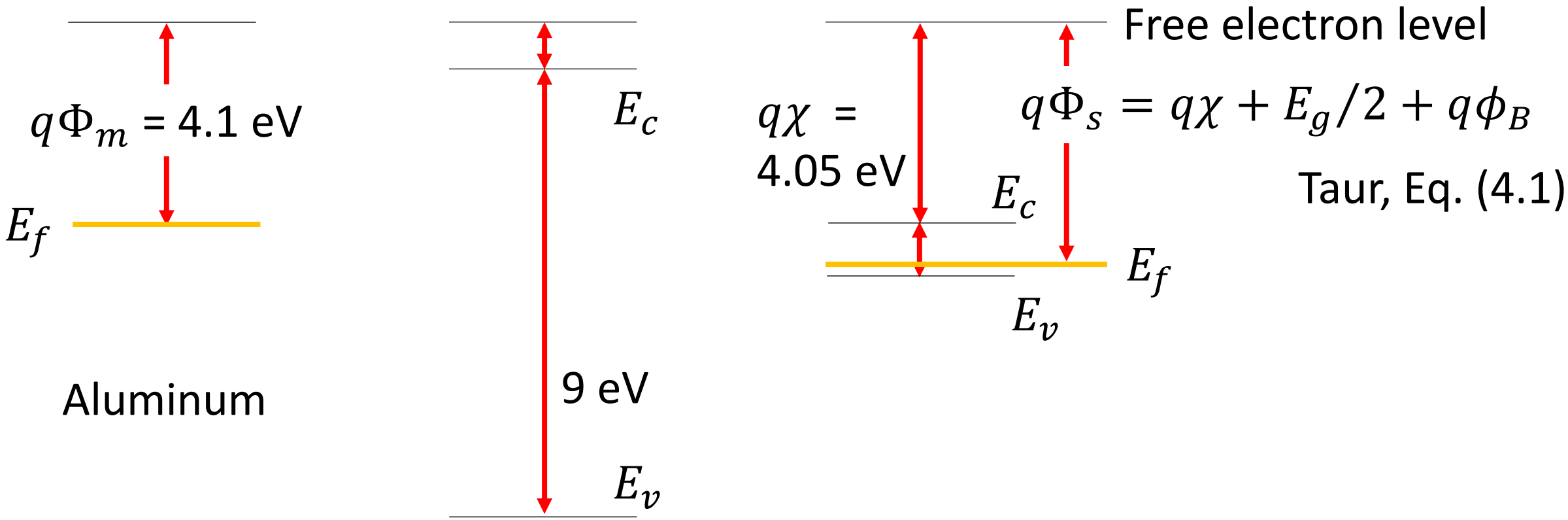
- (001) surface and [110] channel





Energy band diagram

- Three components
  - Metal, silicon dioxide, and p-type silicon

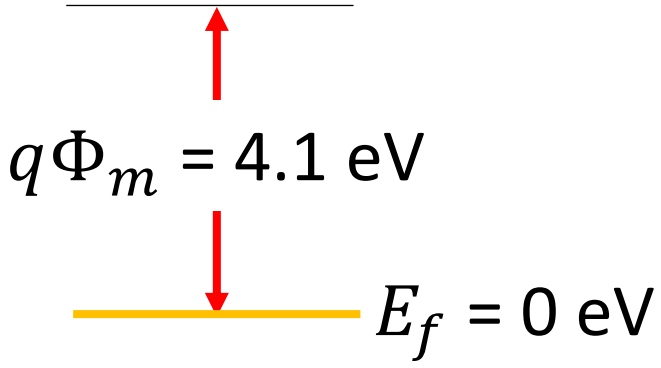


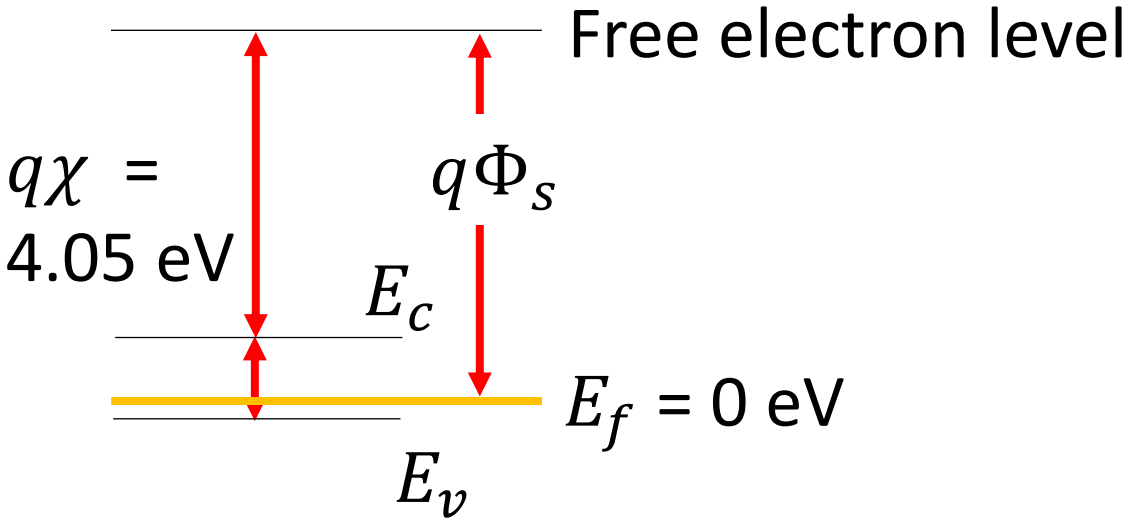
Consider $V_g = V_{sub} = 0$ V.

- Rule: Align the Fermi level.
  - The energy difference is

$$q\Phi_s - q\Phi_m$$

  - It means that a non-zero electric field is applied in the oxide layer.



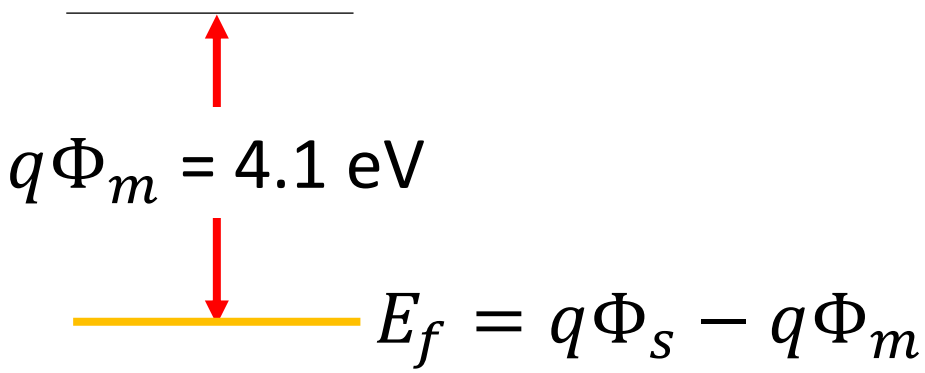


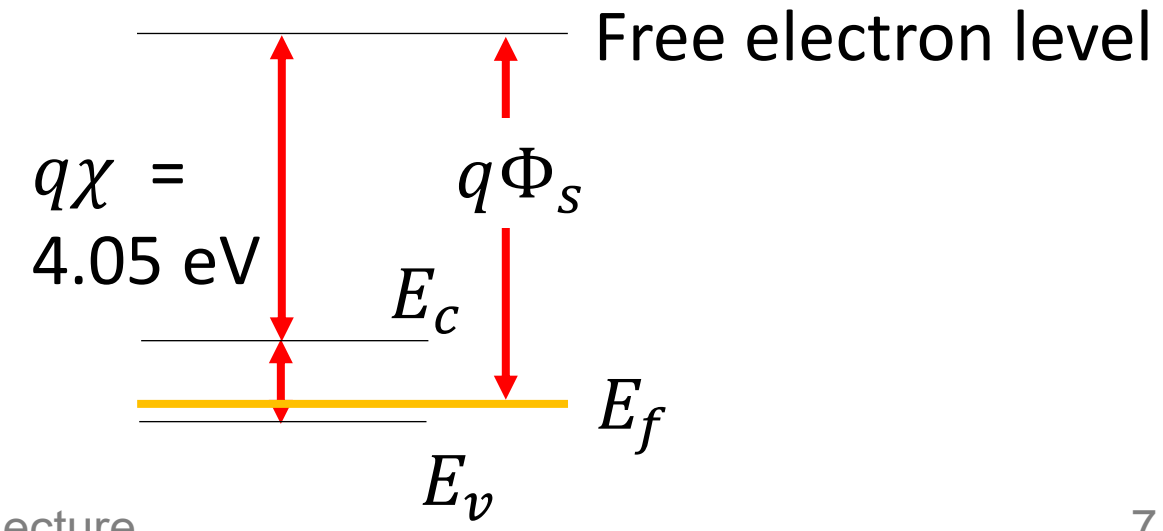
When $V_g = \Phi_m - \Phi_s < 0$,

- The energy band at gate moves upward.
  - There is no energy difference.
  - It means that the energy band becomes flat.
  - This gate voltage is called the flatband voltage, V_{fb}.

$$V_{fb} = \Phi_m - \Phi_s = \Phi_{ms}$$

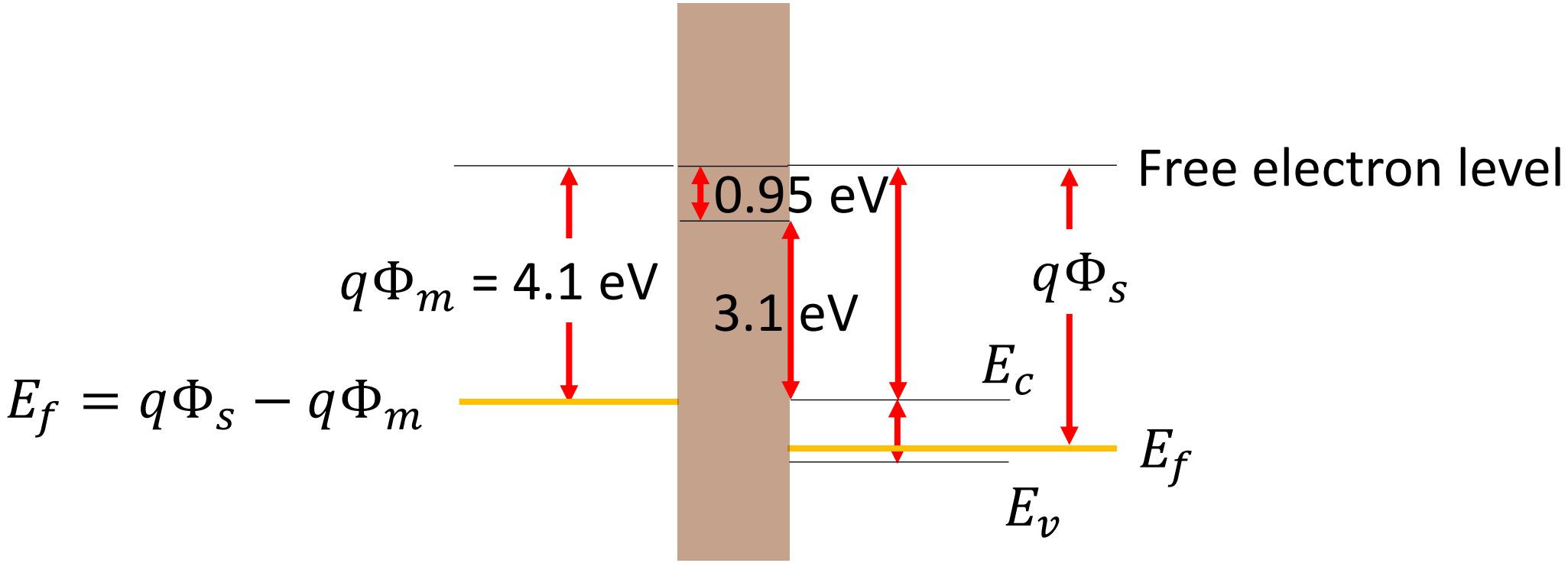
Taur, ~Eq. (4.2)





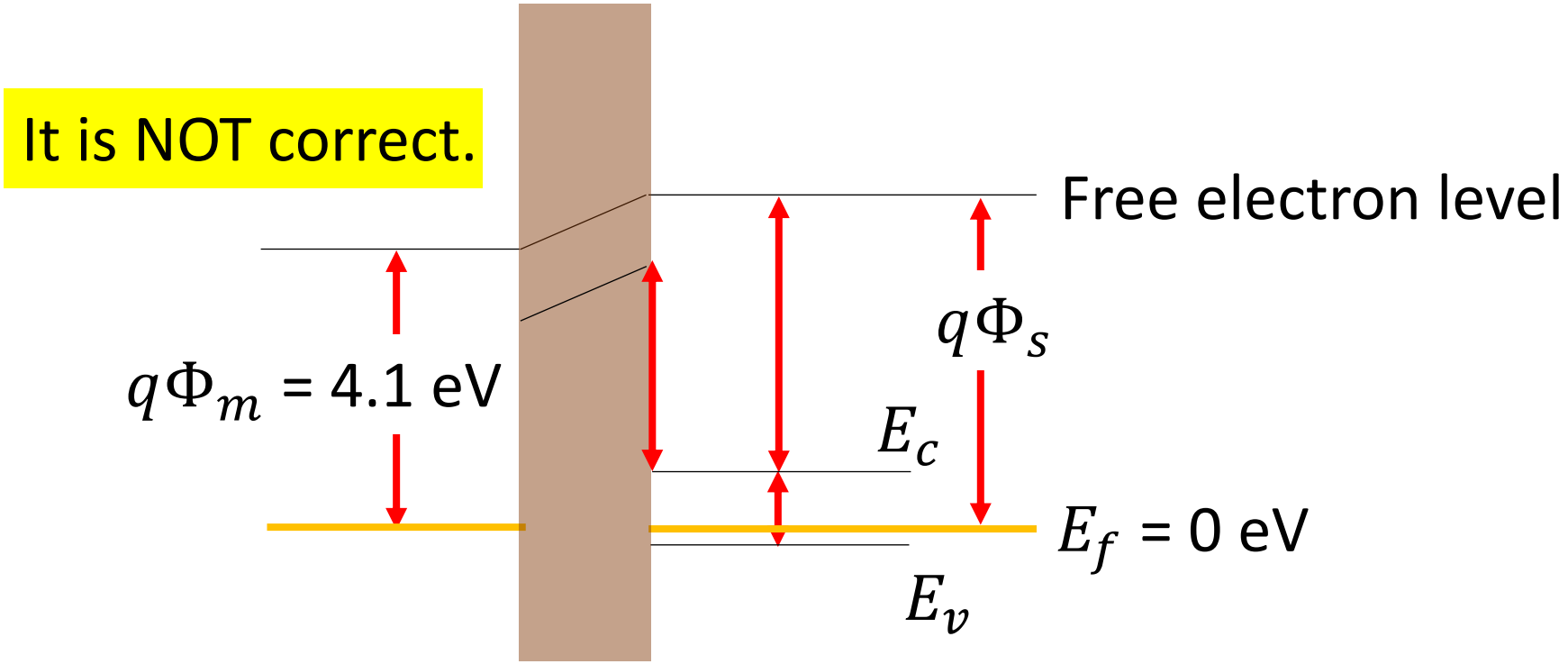
Draw the energy band diagram at $V_g = V_{fb}$.

- Since the energy band is flat, it is not difficult.
  - The electron energy barrier is 3.1 eV between the conduction bands of silicon and silicon dioxide.



Consider $V_g = V_{sub} = 0$ V, again.

- Non-zero electric field is found.
  - However, the energy difference, $q\Phi_s - q\Phi_m$, cannot be solely applied to the oxide layer. *Why?*



Remember the boundary condition.

- At the silicon-oxide interface,

$$\epsilon_{ox}|\mathbf{E}_{ox}| = \epsilon_{si}|\mathbf{E}_{si}|$$

Taur, Eq. (4.6)

  - Therefore, we have a non-zero "surface potential."

$$\phi_s = \phi(0) - \phi(\infty)$$

  - The overall potential difference is $V_g - V_{fb}$.
  - The oxide potential, V_{ox}
  - It is important to note that

Total charge per unit area induced in silicon

$$V_g - V_{fb} = V_{ox} + \phi_s = \frac{-Q_s}{C_{ox}} + \phi_s$$

Taur, Eq. (4.7)

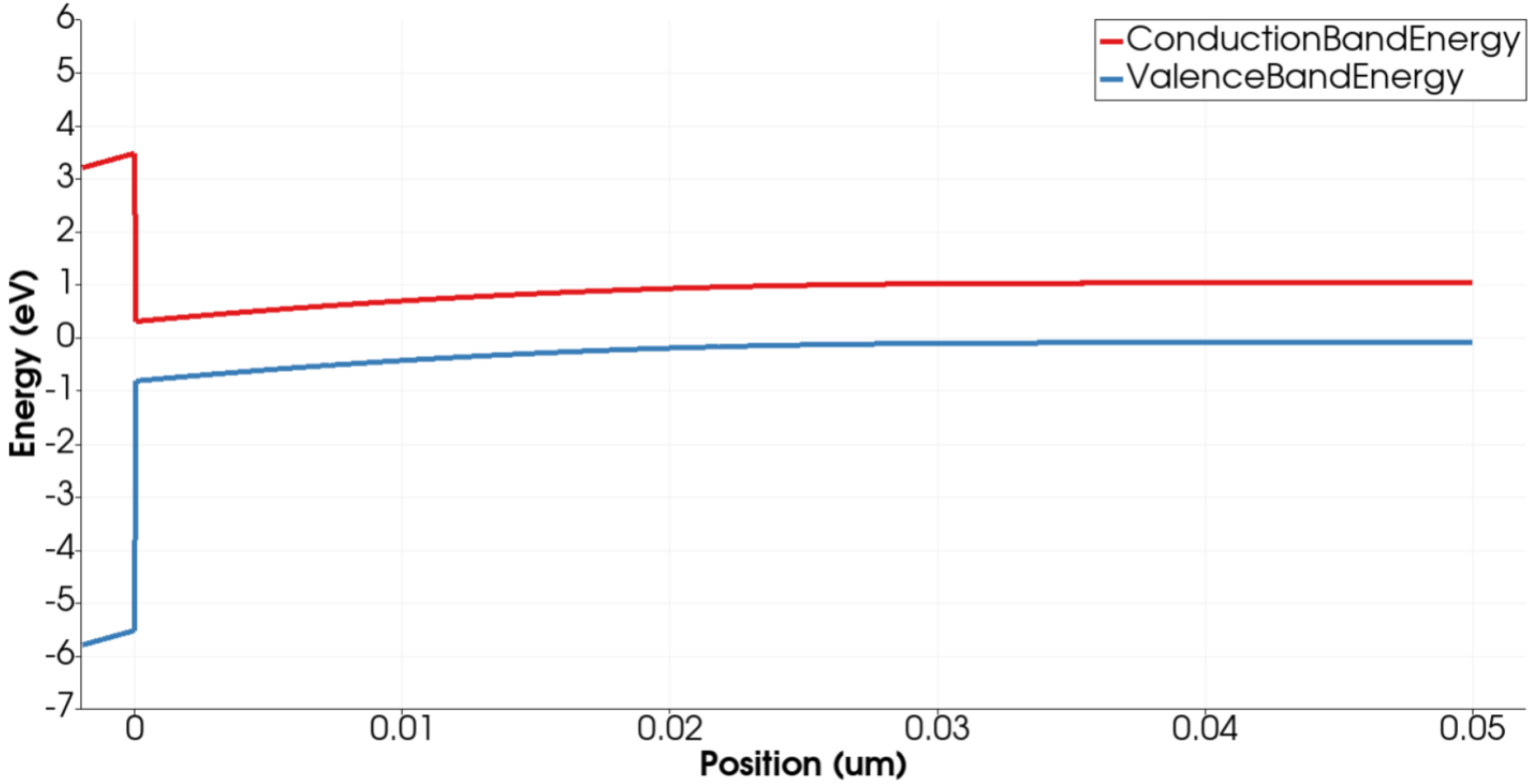
  - C_{ox} is the oxide capacitance per unit area,

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

Taur, Eq. (4.3)

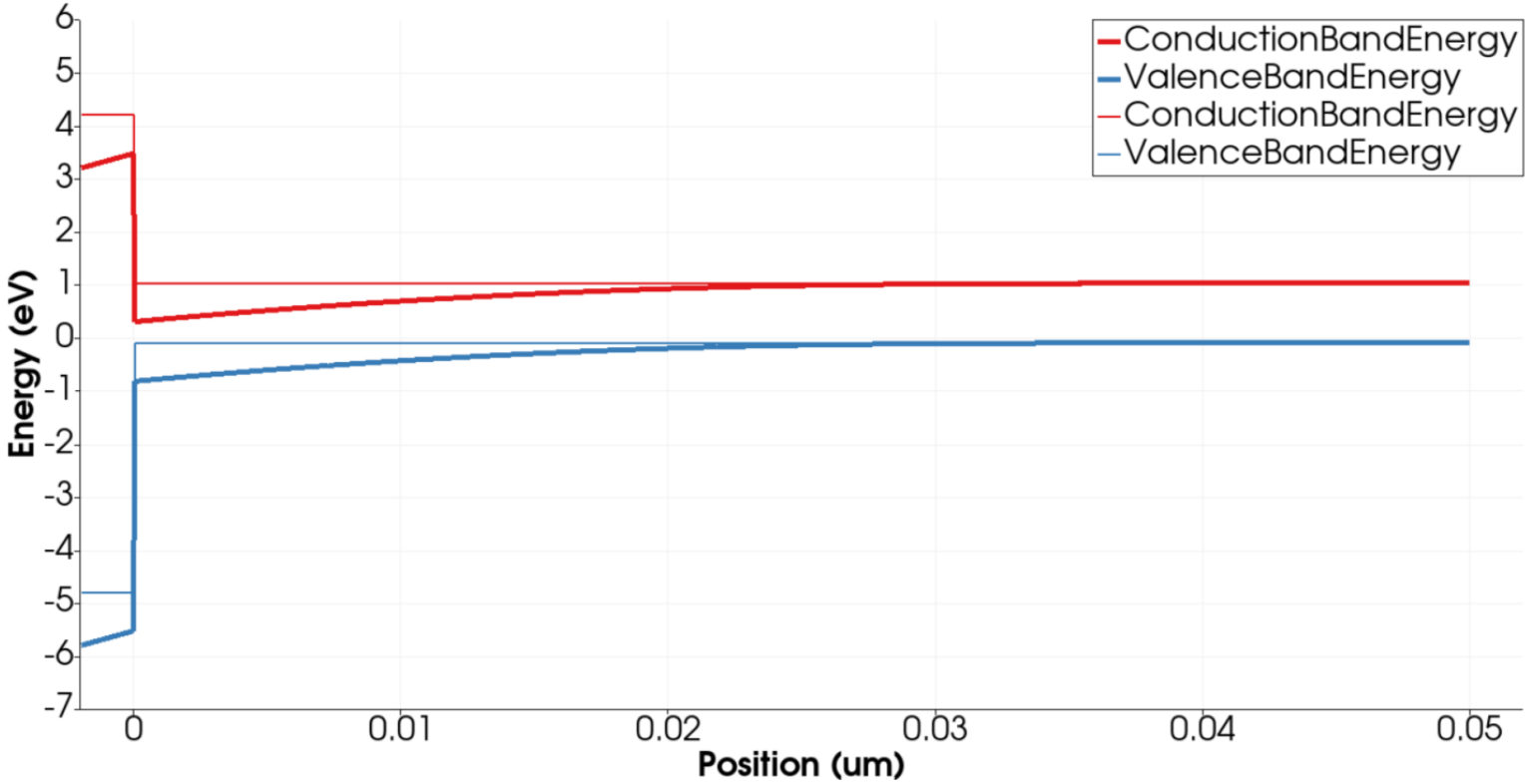
A model MOS @ 0 V

- Gate workfunction = 4.1 eV. 2-nm-thick oxide. 10^{18} cm^{-3}.
  - Depletion



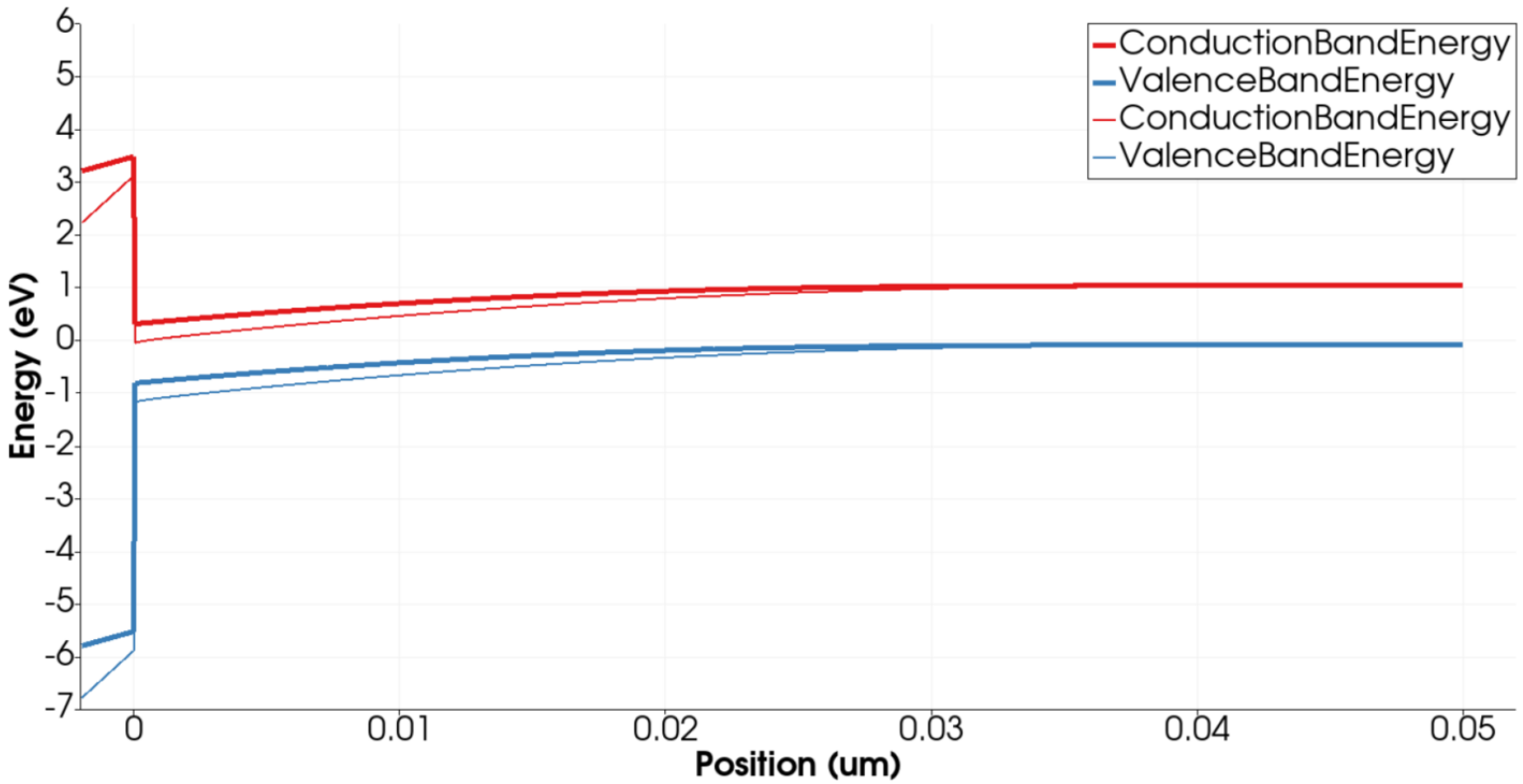
A model MOS @ flatband voltage

- The flatband voltage is -1.01 V.
  - In this case, the band energies are flat.



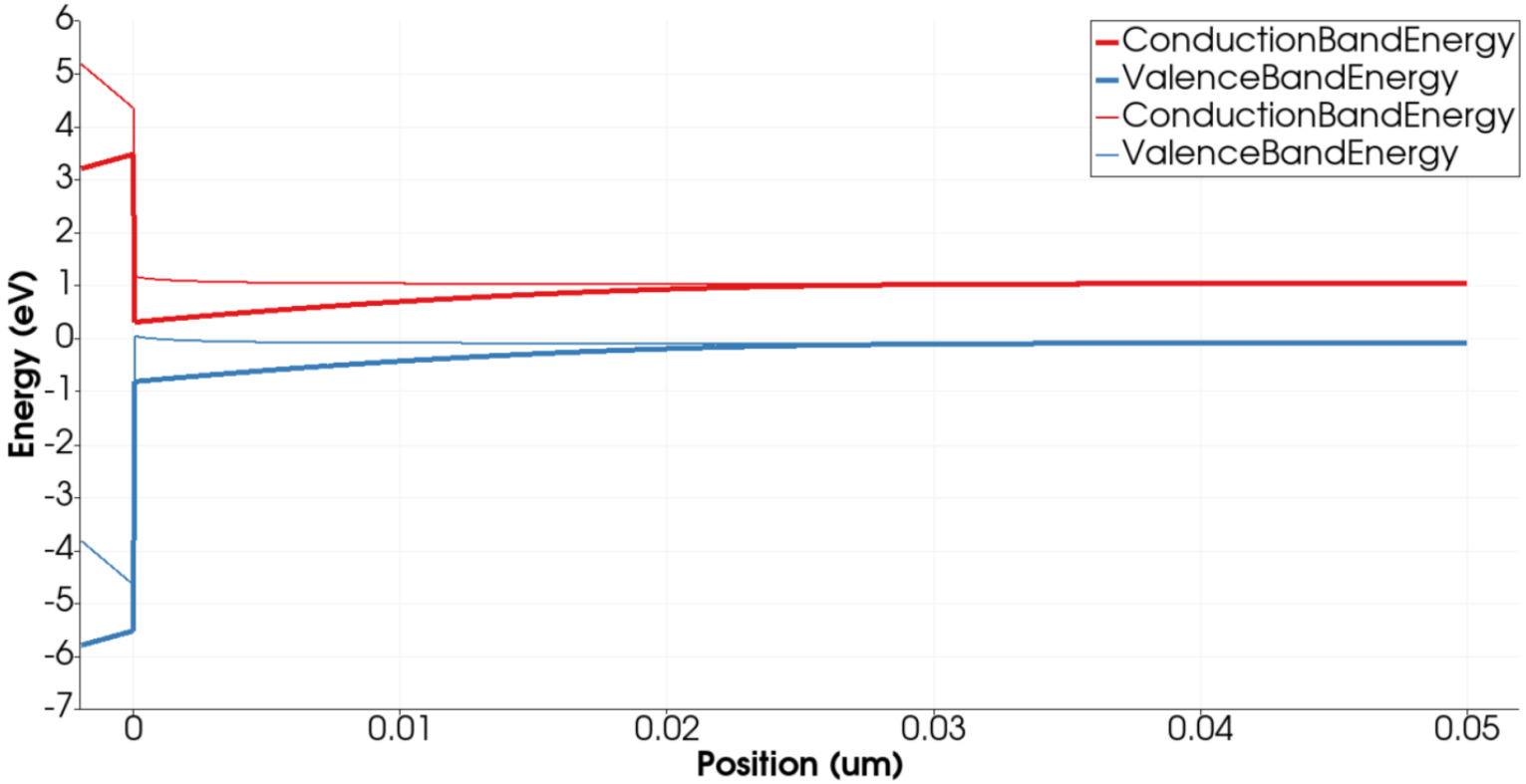
A model MOS @ 1 V

- Further downward band bending
  - Inversion. Increase of surface potential



A model MOS @ -2 V

- Accumulation
  - Upward band bending. A lot of holes



Thank you!